DAPI_PML_SUMO

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- **Developed for:** Adèle
- **Team:** De Thé
- **Date:** October 2025
- **Software:** Jupyter Notebook (for analysis) and Napari (for visualization)

Images description

Images acquired on Zeiss LSM 800 confocal microscope with a 63x objective:

1. 2D images: 1x zoom, pixel size: 0.099 µm (XY)
2. 3D stacks: 2x zoom, voxel size: 0.055 µm (XY), 0.25 µm (Z)

3 channels:

1. DAPI nuclei
2. PML foci
3. SUMO2 or another partner protein

Notebooks description

DAMI_PML_SUMO_3D.ipynb

- Identify focused slices on PML channel using the Laplacian variance method
- Detect nuclei in 3D:
    - Perform maximum-intensity z-projection of DAPI channel
    - Detect nuclei in 2D using Cellpose
    - Extend nuclei into 3D across focused slices
- Detect PML foci in 3D using DoG filtering + Triangle automatic thresholding + opening
- Estimate background intensity in PML and SUMO2 channels as the median pixel intensity outside nuclei
- For each nucleus:
    - Compute the following parameters for each focus: volume, mean and sum of background-corrected intensity in PML channel, mean and sum of background-corrected intensity in SUMO2 channel
    - Compute mean and sum of background-corrected intensity in SUMO2 channel outside PML foci

DAMI_PML_SUMO_2D.ipynb

- Detect nuclei in 2D using Cellpose
- Detect PML foci in 2D using DoG filtering + Triangle automatic thresholding + opening
- Estimate background intensity in PML and SUMO2 channels as the median pixel intensity outside nuclei
- For each nucleus:
    - Compute the following parameters for each focus: area, mean and sum of background-corrected intensity in PML channel, mean and sum of background-corrected intensity in SUMO2 channel
    - Compute mean and sum of background-corrected intensity in SUMO2 channel outside PML foci

Usage instructions

1. Log in to the WS3 workstation in the ORION image analysis room
2. Open the Command Prompt
3. Activate the Conda environment by running:

```
conda activate PML_SUMO
```

5. Launch Jupyter Notebook by running:

```
jupyter notebook
```

6. In your web browser, navigate to and open the `DAPI_PML_SUMO.ipynb` notebook
7. Enter the required parameters in the designated cell
8. Run all cells to perform the data analysis
9. Run the visualization cells to review the results in Napari

Dependencies

- Conda environment **PML_SUMO**
- Cellpose 3 pretrained model `nuclei`

Both the **PML_SUMO** environment and the `nuclei` model are installed on the WS3 workstation in the ORION image analysis room.

To install the environment on another computer, use the `environment.yml` file provided in the repository. If you experience issues installing Cellpose v3.1.1 (preferably the GPU-enabled version), please follow the official installation instructions here: https://github.com/MouseLand/cellpose.

Version history

Version 1 released on October 2, 2025.